Real analysis Lecture 5 and 6 homework

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September 2025

1 Suppose $x, y \in R$ and $x < y$. Prove that there exists $i \in R \setminus Q$ such that $x < i < y$.

Suppose $x, y \in R$ and $x < y$. Therefore $y - x > 0$ and since $\frac{1}{\sqrt{2}} > 0$, therefore $0 < \frac{(y-x)}{\sqrt{2}}$. By the density of the rational numbers, there exists $t \in Q$ such that, $0 < t < \frac{(y-x)}{\sqrt{2}}$. Therefore $0 < t\sqrt{2} < y - x \Rightarrow$

$$x < x + t\sqrt{2} < y$$

. Since t is a rational number, t $\sqrt{2}$ is irrational, setting $i = x + t\sqrt{2}$ the theorem is proved $\square$.

2 Let $E \subset (0, 1)$ be the set of all real numbers with decimal representation using only the digits 1 and 2:

$$E := \{x \in (0,1) : \forall j \in \mathbb{N},\ \exists d_{-j} \in \{1,2\} \text{ such that } x = 0.d_{-1}d_{-2}\ldots\}.$$

Prove that $|E| = |\mathcal{P}(\mathbb{N})|$.

Assume the assumptions in the problem hold true. Let $f : E \longrightarrow \mathcal{P}(\mathbb{N}) = \{j \in \mathbb{N} : d_{-j} = 2\}$ Let there be $f(x_1), f(x_2) \in \mathcal{P}(\mathbb{N})$, where $x_1 = 0.d_1d_2..., x2 = 0.b_1b_2...$, such that $f(x_1) = f(x_2) \Rightarrow \forall j \in N | d_j = b_j \Rightarrow x_1 = x_2 \Rightarrow$

$$E \geq |\mathcal{P}(\mathbb{N})|$$

Assume $x_1 = x_2 \Rightarrow 0.d_1d_2... = 0.b_1b_2... \Rightarrow \forall j \in N | d_j = 2 : b_j = 2 \Rightarrow \forall n \in f(x_1)\ \exists n \in f(x_2) \Rightarrow f(x_1) = f(x_2) \Rightarrow$

$$|\mathcal{P}(\mathbb{N})| \geq |E|$$

$$|\mathcal{P}(\mathbb{N})| = |E| \qquad \square$$

3 Let A and B be two disjoint, countably infinite sets. Prove that $A \cup B$ is countably infinite.

Since $|A| = |\mathbb{N}|$ and $|A \cup B| \geq |A|$, therefore $|A \cup B| \geq |\mathbb{N}|$. Let $A \cup B = \{a_1, a_2, ..., b_1, b_2, ...\} = \{x_1, x_2, ...\}$ and $f : A \cup B \longrightarrow \mathbb{N}$

$$f(x_n) = n$$

trivially, $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \Rightarrow |A \cup B| \leq |A| \qquad \square$.

4 Prove that the set of irrational numbers, $\mathbb{R}\backslash\mathbb{Q}$, is uncountable. You may use the facts discussed in the lectures that $\mathbb{R}\backslash\mathbb{Q}$ is infinite and $\mathbb{R}$ is uncountable without proof.

Assume $\mathbb{R}\backslash\mathbb{Q}$ is countable. Since $\mathbb{Q}$ is countable by the previous theorem, their union is countable. Therefore

$$\mathbb{R}\backslash\mathbb{Q} \cup \mathbb{Q} = \mathbb{R}$$

is countable $\longrightarrow\!\longleftarrow$.

5 Let $A \subset \mathbb{R}$ be bounded above, and let a_0 be an upper bound for A. Prove that $a_0 = \sup A$ if and only if for every $\varepsilon > 0$, there exists $a \in A$ such that

$$a_0 - \varepsilon < a.$$

Let $A \subset \mathbb{R}$ and $\varepsilon > 0$, assume $a_0 = supA$; therefore,, for all a $\in A$ $a_0 > a$. Assume $\exists a \mid \varepsilon - a_0 \geq a$, then that would imply a is not the supremum of A. Therefore, for every $\varepsilon > 0$, there exists $a \in A$ such that $a_0 - \varepsilon < a$..

6 Prove that

$$\lim_{n \to 0} \frac{1}{20n^2 + 20n + 2020} = 0$$

Choose $M \in N$ such that $\frac{1}{20M^2} < \epsilon$, so for all $n \geq M$.

$$|\frac{1}{20n^2 + 20n + 2020} - 0| = \frac{1}{20n^2 + 20n + 2020} \leq \frac{1}{20n^2} \leq \frac{1}{20M^2} \leq \epsilon$$